HIGH RESOLUTION NIGHT TIME REMOTE SENSING FOR NATURAL DISASTER MONITORING

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by

Priyanshi Agarwal - 2022BCS0186

Shaik Ali - 2022BCS0118

More Ojas Ganesh - 2022BCD0043

Kalabandi Karthik Raj - 2022BCD0041

to

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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DECLARATION

We, **Priyanshi Agarwal**, **Shaik Ali**, **More Ojas Ganesh**, **Kalabandi Karthik Raj** hereby declare that, this report entitled **High resolution night time remote sensing for natural disaster monitoring** submitted to Indian Institute of Information Technology Kottayam towards partial requirement of Bachelor of Technology in Computer Science is an original work carried out by us under the supervision of **Dr.Priyadharshini S** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. We have sincerely tried to uphold the academic ethics and honesty. Whenever an external information or statement or result is used then, that have been duly acknowledged and cited.

Kottayam-686635 **Priyanshi Agarwal (2022BCS0186)**

November 2025 **Shaik Ali (2022BCS0118)**

More Ojas Ganesh (2022BCD0043)

Kalabandi Karthik Raj (2022BCD0041)

CERTIFICATE

This is to certify that the work contained in this project report entitled **"High resolution night time remote sensing for natural disaster monitoring"** submitted by **Priyanshi Agarwal**, **Shaik Ali**, **More Ojas Ganesh**, **Kalabandi Karthik Raj** to the Indian Institute of Information Technology Kottayam towards partial requirement of Bachelor of Technology carried out by them under my supervision and that it has not been submitted elsewhere for the award of any degree.

Kottayam-686635 Dr.Priyadharshini S

November 2025 Project Supervisor

ABSTRACT

Timely and accurate detection of humans and flood-affected regions during natural disasters is critical for efficient rescue operations and resource allocation. This work focuses on developing and evaluating high-resolution night-time remote sensing techniques for natural disaster monitoring using both optical and Synthetic Aperture Radar (SAR) data. Two distinct datasets were utilized: the **C2A dataset**, containing synthetic disaster imagery representing humans in flood scenarios, and the **SEN12-FLOOD dataset**, which provides co-registered Sentinel-1 SAR and multispectral data for flood detection tasks. A **YOLO-based human detection model** was trained on the C2A imagery to identify trapped or stranded individuals. For SAR-based flood monitoring, a **Convolutional Neural Network (CNN)** was implemented. Building upon this, a **Vision Transformer (ViT)** model was trained on the same Sentinel-1 SAR dataset, achieving improved feature representation and accuracy. The results demonstrate that combining high-resolution optical imagery with SAR-based sensing techniques enables complementary understanding of disaster scenes.

Contents

List of Figures **viii**

List of Tables **ix**

1 Introduction **1**

1.1 Motivation 2

1.2 Challenges 4

1.3 Objectives 6

2 Related Work **8**

2.1 Night-Time Disaster Imaging and Sensing 8

2.2 Deep Learning and Vision Transformers for Low-Light Scene Understanding 9

2.3 AI-Driven Decision Support and Agentic Coordination Systems 10

2.4 Summary and Research Gap 11

3 Proposed Methodology **14**

3.1 SAR Image Acquisition 14

3.2 Preprocessing and Feature Normalization 17

3.3 ViT Perception Module 18
3.3.1 Patch Embedding 18
3.3.2 Positional Encoding 19
3.3.3 Transformer Encoder Layers 19
3.3.4 Disaster Feature Extraction 20
3.4 Structured Disaster Representation 22
3.5 Human Detection via YOLOv8 on C2A Dataset 22
3.6 LangGraph Multi-Agent System 23
3.6.1 RAG (Retrieval-Augmented Generation) Agent 24
3.6.2 Reasoning & Planning Agent 24
3.6.3 Routing & Logistics Agent 25
3.6.4 Communication Agent 26
3.7 Rescue Team Interface 27
3.8 Feedback Loop 28

4 Experimentation Results 29
4.1 Human Detection on C2A Dataset 29
4.1.1 Confusion Matrix 29
4.1.2 Loss Curves 30
4.1.3 Recall and Precision 33
4.1.4 Summary 34
4.2 Flood Classification on SEN12-FLOOD Dataset 34
4.3 Overall Analysis 38

5 Conclusion and Future Work 39

Bibliography **42**

List of Figures

3.1 Unified workflow 15

4.1 Confusion matrix of YOLOv8 on the C2A dataset 30
4.2 Training loss curves 31
4.3 CNN/ViT metrics over epochs. 32
4.4 Precision and recall curves 33
4.5 CNN training and validation loss on the SEN12-FLOOD dataset 35
4.6 CNN training and validation accuracy 35
4.7 CNN training and validation F1-score 36
4.8 Confusion matrix for ViT 37
4.9 SAR flood detection outputs 37

List of Tables

4.1 Classification performance 34
4.2 CNN performance 34
4.3 ViT Classification Report 36

Chapter 1

Introduction

Natural disasters such as floods, fires, and building collapses pose serious threats to human life and infrastructure. Accurate and timely detection of affected regions and trapped individuals is essential for efficient rescue operations, yet achieving reliable results during night-time or low-visibility conditions remains challenging. Conventional optical remote sensing systems often fail under poor illumination or atmospheric interference, whereas **Synthetic Aperture Radar (SAR)** sensors provide valuable all-weather, day-and-night imaging capability.

Recent progress in deep learning has significantly advanced disaster monitoring. **Convolutional Neural Networks (CNNs)** have been widely used for flood detection and land-cover analysis in SAR data, though their limited ability to capture long-range spatial dependencies restricts performance in complex scenes. To overcome this, **Vision Transformers (ViT)** leverage self-attention mechanisms to model global context and have shown superior results on high-resolution remote sensing datasets such as Sentinel-1.

Meanwhile, the **YOLO** family of object detectors enables real-time human detection from optical imagery, making it highly suitable for UAV-based disaster assessment. Integrating both optical and SAR-based approaches creates a robust framework for **high-resolution night-time disaster monitoring**. In this study, three models were explored: YOLO for human detection on the C2A imagery dataset [1], CNN for flood detection using the SEN12-FLOOD SAR dataset [2], and ViT for improved SAR-based flood mapping. Together, these approaches highlight the potential of combining multi-modal remote sensing and deep learning for scalable, real-time disaster response.

1.1 Motivation

Night-time post-disaster scenarios present extreme challenges for search and rescue teams due to poor visibility, unpredictable environmental conditions, and limited accessibility of affected areas. Under such circumstances, locating trapped victims or identifying flood-affected regions quickly and accurately becomes vital for saving lives and optimizing resource deployment. The motivation for this work stems from the need to leverage advanced remote sensing and deep learning techniques to automate these critical detection processes across different sensing domains — optical and radar-based.

- **Low Visibility and Complex Conditions:** Manual observation becomes highly unreliable during darkness, smoke, fog, or cloud-covered scenarios. Human operators often miss important cues in such visually degraded environments. Deep learning models such as YOLO can

enhance detection accuracy by identifying humans and objects even in poor lighting. These models improve contrast, recognize outlines, and track movement patterns effectively. As a result, they significantly support night-time and low-visibility rescue operations.

- **Need for All-Weather Flood Monitoring:** Flood detection during heavy rainfall or night-time is difficult using optical sensors due to limited lighting and cloud obstruction. SAR data solves this challenge as it penetrates clouds and functions independently of illumination conditions. When combined with CNN or ViT-based architectures, SAR imagery enables accurate water body detection and terrain mapping. This ensures continuous monitoring regardless of weather or time. Thus, SAR-based approaches make flood assessment more reliable and uninterrupted.

- **Time-Critical Rescue Response:** Traditional search operations are time-consuming and often delay rescue actions, reducing survival rates. Automated deep learning systems can process large volumes of visual and radar data within seconds. These models quickly identify trapped individuals, hazardous areas, and safe paths for teams. Faster detection supports coordinated and efficient rescue decision-making. Overall, automation enhances response speed and effectiveness during critical emergencies.

- **Safety of Rescue Teams:** Rescue teams frequently face dangerous environments such as unstable buildings, deep floodwaters, or contaminated regions. Automated analysis of imagery and SAR data allows

the assessment of these zones remotely without risking personnel. Such systems detect structural damage, estimate flood levels, and locate inaccessible areas. This remote evaluation minimizes direct human exposure to hazardous conditions. Consequently, it improves both operational safety and preparedness.

- **Scalable and Intelligent Solutions:** AI-driven frameworks can be deployed on drones, satellites, ground robots, and fixed sensor systems for large-scale monitoring. These scalable solutions provide real-time analytics and automated alerts across wide disaster-affected regions. They handle diverse sensor inputs and adapt to varying environmental conditions. Intelligent models ensure continuous tracking of disaster progression without requiring large manpower. This makes the entire monitoring system more efficient and robust.

In summary, this project is motivated by the urgent need to develop an intelligent, dual-domain system that integrates optical and SAR remote sensing for comprehensive night-time disaster monitoring. By combining YOLO-based human detection on the C2A imagery with CNN and Vision Transformer models for flood detection on the SEN12-FLOOD dataset, the system aims to enhance detection accuracy, operational safety, and real-time efficiency in post-disaster response and management.

1.2 Challenges

Developing an automated night-time disaster monitoring system using both optical and radar data presents several technical and practical challenges.

The diversity of sensor modalities, environmental conditions, and computational requirements introduces complexity at every stage—from dataset preparation to model deployment. The key challenges encountered in this project are outlined below:

- **Low-Visibility and Illumination Variations:** Night-time images in the C2A dataset [2] exhibit poor lighting, motion blur, and high contrast differences, making it difficult for models to distinguish humans from background clutter.

- **Cross-Domain Data Heterogeneity:** The project integrates two fundamentally different data modalities—optical RGB imagery (C2A) and radar-based SAR imagery (SEN12-FLOOD). The differing spatial resolutions, noise patterns, and feature representations make unified model training and comparison non-trivial.

- **Limited Availability of Night-Time Data:** Real-world night-time disaster datasets are scarce, forcing reliance on synthetic data and augmentation techniques to simulate low-light conditions. This may lead to a domain gap between training and real-world deployment.

- **Noise and Speckle in SAR Images:** SAR data inherently contains speckle noise, which degrades visual quality and complicates feature extraction.

- **Small Object Detection:** In optical disaster scenes, humans often appear as small or partially occluded objects, posing a significant chal-

lenge for detectors like YOLO, which struggle with tiny object localization.

Overall, these challenges highlight the inherent complexity of combining optical and SAR-based remote sensing for real-time, night-time disaster monitoring. Addressing these limitations through improved data preprocessing, model optimization, and domain adaptation techniques forms a crucial direction for ongoing and future research.

1.3 Objectives

The primary objective of this project is to design and evaluate a deep learning–based framework for night-time disaster monitoring using both optical and radar remote sensing data. The system aims to automate the detection of humans and flood-affected regions in challenging conditions, thereby improving the speed, accuracy, and safety of post-disaster response operations. The specific objectives of this work are outlined below:

- **Human Detection in Night-Time Imagery:** Train and evaluate YOLOv8 model on the C2A dataset [1] to accurately detect humans in disaster scenarios under low-light conditions.

- **Flood Detection Using SAR Data:** Develop and implement a Convolutional Neural Network (CNN) for classifying flooded and non-flooded regions using Sentinel-1 SAR data from the SEN12-FLOOD dataset [2].

- **Transformer-Based Performance Enhancement:** Employ a Vision Transformer (ViT) model on the SEN12-FLOOD dataset to capture long-range spatial dependencies and evaluate improvements over the CNN baseline.

- **Cross-Modal Analysis:** Compare the performance of optical-based and radar-based approaches to demonstrate the complementary benefits of integrating multi-modal remote sensing for disaster monitoring.

In summary, this project aims to build a dual-domain deep learning framework capable of detecting humans and assessing flood impacts in nighttime disaster environments. By leveraging YOLO for optical imagery and Transformer-enhanced models for radar data, the system contributes toward scalable, intelligent, and real-time disaster management solutions.

Chapter 2

Related Work

This chapter examines the body of research on AI-driven emergency management decision support systems, developments in low-light image enhancement, and nighttime disaster monitoring using satellite and aerial imagery. The research gaps that drive the suggested high-resolution, AI-assisted nighttime disaster management framework are also identified.

2.1 Night-Time Disaster Imaging and Sensing

Because of atmospheric scattering and low signal-to-noise ratios, conventional remote sensing systems are severely limited in low-light or nighttime conditions. The advent of VIIRS (Visible Infrared Imaging Radiometer Suite) and SWIR (Short-Wave Infrared) sensors has made it possible to monitor fire outbreaks, flood levels, and power outages at night with coarse-grained information.[3]

However, these sensors' 375 m or coarser spatial resolutions are insufficient for responding to local or urban disasters. Improved feature visibility for object and structure recognition during nighttime imaging has been demonstrated by recent studies employing low-light image enhancement techniques (e.g., RetinexNet, Zero-DCE).

Researchers have used drone-based thermal imaging and multispectral satellite data to improve spatial detail, which makes it possible to identify survivors and damaged areas more precisely in post-disaster situations. However, these methods still lack scalable deployment in emergency situations and automated interpretation.

2.2 Deep Learning and Vision Transformers for Low-Light Scene Understanding

Recent advances in Vision Transformers (ViT) and hybrid CNN–Transformer models have shown strong capability in understanding complex, low-light environments. Because of their global attention mechanism and capacity to learn from sparsely labeled data, ViT-based models outperformed classical CNNs in studies like Zhou et al. (2023) that employed them for flood detection and fire segmentation.

Furthermore, to enhance visual perception in extremely low light levels, nighttime object detection frameworks utilizing DarkViT, YOLOv8-Night, and LLFormer architectures have been developed.

These findings imply that, even in visually impaired situations, such as those that occur during nighttime disasters, combining ViT-based vision

models with real-time satellite or drone feeds can significantly increase detection accuracy.

2.3 AI-Driven Decision Support and Agentic Coordination Systems

Early warning, risk assessment, and post-disaster logistics have all benefited from the successful application of AI-based decision support systems (DSS). For example, reinforcement learning agents and graph-based reasoning models have been used to optimize resource allocation and evacuation routes.

Existing DSS frameworks, however, are still compartmentalized; they do not have contextual awareness across various data sources or multi-agent coordination.

A promising solution is provided by the new paradigm of AI agents and frameworks like **LangGraph**. LangGraph makes it possible to create self-governing, intelligent agents that are capable of:

Combining multimodal data (inputs from social media, sensors, drones, and satellites) This entails combining data from various sources, including satellite imagery, drone footage, ground sensors, and social media posts. Situational awareness at both the macro and micro levels is produced by combining these inputs. Satellites record large-scale environmental conditions, drones provide local high-resolution visuals, social media provides real-time human observations, and sensors provide precise measurements. The system minimises blind spots and enhances the precision and promptness of insights by integrating these modalities.

Planning activities together (e.g., risk prioritization, resource dispatch) This is the coordination of actions like risk assessment, critical area prioritisation, and resource deployment planning within a single workflow. The system assesses threat levels, prioritises areas that need to be addressed right away, and suggests the best way to deploy teams or equipment. Planning together reduces delays and avoids effort duplication. In the end, it increases response efficiency by ensuring that decisions are data-driven, consistent, and in line with operational goals.

Giving human decision-makers comprehensible chains of reasoning This ensures that system recommendations are transparent and easily interpretable by human operators. Instead of providing black-box outputs, the system explains how conclusions were reached, showing the supporting evidence and intermediate reasoning steps. These explanations build trust, allow decision-makers to validate outputs, and support corrections if needed. Clear reasoning also helps teams with varying technical backgrounds collaborate more effectively with AI systems.

Therefore, combining ViT-driven perception modules with LangGraph-based AI agents may allow for a completely automated disaster response pipeline, from understanding scenes in real time to developing and implementing strategies.

2.4 Summary and Research Gap

Even though nighttime imaging, low-light enhancement, and disaster detection have made significant strides, no end-to-end AI system presently inte-

grates:

The system includes a ViT-based perception module that can process high-resolution satellite and drone images even at night or in low light. The ViT uses global attention mechanisms to extract structural patterns and fine details from noisy optical, thermal, and SAR data. When visibility is poor, this makes it possible to accurately detect changes in the terrain, the spread of floods, and human presence. In a variety of imaging environments, the model guarantees consistent scene understanding. Consequently, it offers trustworthy perception outputs for tasks involving downstream decision-making.

To create efficient disaster response plans, AI-powered decision-making agents cooperate within an inter-agent coordination framework. To create cogent action plans, these agents combine perception results, GIS data, historical reports, and environmental indicators. The coordination layer maintains synchronisation between the resource allocation, planning, and routing modules. Decisions are dynamically updated in response to fresh observations or field information. This multi-agent strategy significantly improves responsiveness and operational efficiency.

A user-friendly dashboard that offers real-time updates on risk levels, identified individuals, and suggested rescue routes is provided to field responders. Teams can swiftly comprehend changing circumstances during emergencies thanks to the interface's straightforward design. Integration of satellite updates, AI-generated insights, and on-the-ground reports is ensured by continuous data refresh. Additionally, the dashboard has mobile accessibility for field deployment, alerts, and map overlays. Responders can make prompt and informed decisions with the aid of these capabilities.

By creating a multi-agent framework driven by LangGraph and combining context-aware decision support with ViT-enhanced visual intelligence, the suggested system seeks to close this disparity. This will make it possible to detect and manage disasters in real time at night with human-interpretable results.

Chapter 3

Proposed Methodology

The proposed system integrates nocturnal human detection with synthetic aperture radar (SAR)-based disaster perception to provide real-time situational awareness, support rescue planning, and generate actionable intelligence during disaster events. The methodology combines deep learning, multi-agent reasoning, and geographic information system (GIS)-based routing to produce end-to-end disaster response recommendations. The system architecture is depicted in Figure 3.1.

The system is designed to handle both optical (night-time human detection) and radar-based (SAR flood, landslide, and structural damage) inputs, providing a unified framework for disaster monitoring and rescue operations.

3.1 SAR Image Acquisition

Synthetic Aperture Radar (SAR) satellites use microwave signals that can pass through smoke, rain, and clouds to obtain back scatter images of areas

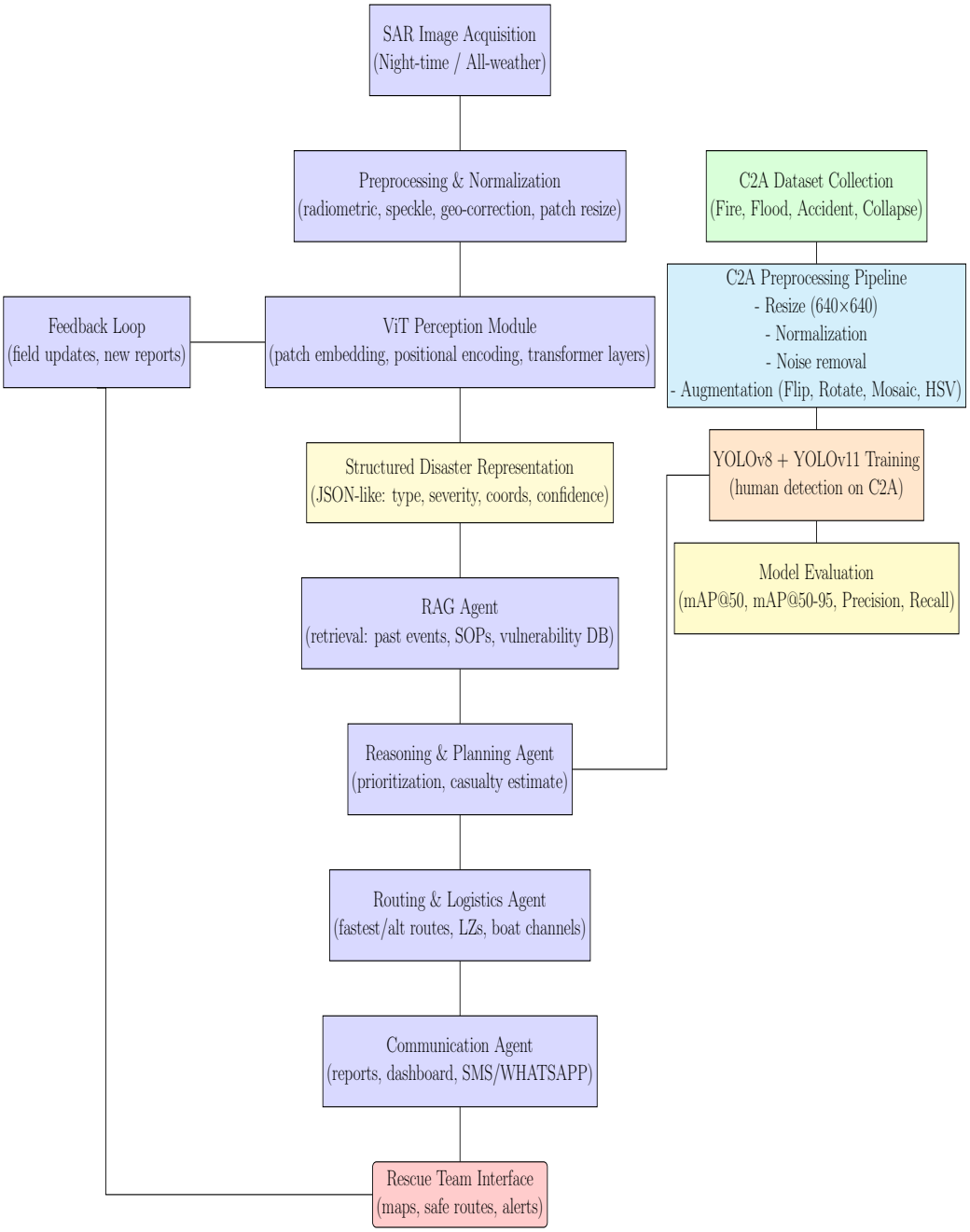

Figure 3.1: *Unified workflow*

affected by disasters. Because of this feature, SAR is especially dependable in extreme weather situations, when conventional optical sensors frequently malfunction. SAR data allows for a thorough examination of surface conditions because radar signals interact differently with water, soil, and damaged infrastructure. Additionally, pre- and post-disaster imagery can be obtained through continuous monitoring, giving downstream AI models a solid observational basis.

SAR imaging is an essential tool for emergency response operations because it is still effective at night and in cloudy or smoky conditions. In contrast to optical sensors that rely on sunlight, SAR systems actively emit their own radar pulses, ensuring consistent imaging regardless of ambient light or atmospheric visibility. Response organisations are able to access continuous, current disaster maps as a result. This dependability facilitates quicker, better-informed decision-making and greatly improves early situational awareness.

Important characteristics like water extent, structural damage patterns, and terrain deformation are all captured by SAR data and are crucial for disaster analytics. While collapsed buildings and displaced structures show unique radar signatures, variations in back scatter intensity can be used to identify flooded areas and landslide-prone surfaces. In addition to providing high-quality input for predictive modeling pipelines, such as Vision Transformer (ViT)-based perception modules, this rich geospatial data facilitates the creation of precise hazard maps.

3.2 Preprocessing and Feature Normalization

Radiometric calibration is applied to remove sensor biases and inconsistencies across SAR acquisitions. It ensures the pixel intensities represent true backscatter values rather than sensor noise. This makes the data more reliable for machine learning models. Calibration also standardizes data collected from different satellites. The process improves both accuracy and interpretability.

Speckle noise filtering is performed to enhance SAR image clarity, as raw radar images naturally contain granular interference. Filters such as Lee or Frost reduce noise while preserving critical structural edges. This step improves the visibility of water bodies, buildings, and terrain features. Removal of speckle improves downstream feature extraction. The filtering ensures the model receives cleaner visual information.

Geometric correction aligns SAR images with geographical coordinates, removing distortions caused by satellite angle or terrain height. This ensures every pixel corresponds to its exact physical location on the map. Accurate alignment enables better integration with GIS layers and optical imagery. The correction step is crucial for multi-modal fusion with ViT and YOLO outputs. It strengthens geospatial accuracy in risk assessment.

After corrections, images are resized into fixed-size patches such as 256×256 to match the ViT input requirements. Patch-based processing allows the transformer to analyze local and global patterns effectively. Resizing ensures uniformity across the dataset. This aids in stable training and reduces computational load. The process also enables batch-

wise processing.

The final output of preprocessing is a clean, normalized batch of SAR image patches ready for the ViT perception pipeline. The standardized format ensures consistent feature extraction across the dataset. These refined inputs reduce model confusion and accelerate convergence. Normalized patches also improve detection quality for flood masks and terrain changes. This preprocessing stage strengthens the overall accuracy of the entire system.

3.3 ViT Perception Module

The Vision Transformer (ViT) serves as the core perception model responsible for extracting high-level disaster features from SAR images. Its attention-based architecture enables global pattern understanding, making it highly effective for identifying complex flood patterns, terrain shifts, and structural deformations. ViT analyzes SAR data both locally and contextually, offering superior feature richness compared to conventional CNNs. The following subsections describe the components in detail.

3.3.1 Patch Embedding

SAR images are divided into fixed-size patches such as 16×16 or 32×32. Each patch acts as a small local region that the transformer processes independently. This division converts the 2D image into a sequence-like representation. The patching step helps the model focus on localized textures like water surfaces or rubble patterns. It forms the fundamental input structure

for the ViT architecture.

Each patch is projected into a D-dimensional embedding using a learnable linear projection. This transforms raw pixel values into a dense numerical representation the model can understand. The projection maps spatial features into a uniform latent space. It enhances the ability of the model to compare and relate distant regions. These embeddings form the starting tokens for the transformer encoder.

3.3.2 Positional Encoding

To preserve spatial information lost during patch extraction, positional embeddings are appended to each patch embedding. The model cannot comprehend the relative location or order of patches without this. Both absolute and relative positions are encoded in the embeddings. This aids ViT in distinguishing between structural layouts and terrain boundaries. In order to reconstruct spatial coherence in SAR imagery, they are crucial.

The model can comprehend water spread dynamics, deformational patterns, and terrain structure by integrating positional data. It improves awareness of worldwide trends like expanding landslides or linked flood zones. The transformer's ability to meaningfully interpret SAR clusters is ensured by positional encoding. This results in more precise segmentation and mapping of disasters. Overall, there is a notable improvement in spatial awareness.

3.3.3 Transformer Encoder Layers

- Multi-Head Self-Attention (MHSA) enables the model to capture global relationships across patches, such as widespread flooding or distant yet

related structural collapses. Each attention head focuses on different aspects of the SAR texture. This multi-view understanding helps the model link cause-and-effect patterns. It is particularly useful for identifying large-scale disaster impacts. MHSA is the core driver of ViT's global reasoning ability.

- Feed-forward networks extract deeper and more abstract SAR textures after the attention layers. They refine the features by processing each token independently. This two-step combination enhances robustness against noisy or subtle SAR patterns. FFNs help distinguish between water surfaces, soil changes, and debris. They significantly increase model expressiveness.
- Layer normalization and skip connections are included to stabilize training and prevent gradient degradation. Skip connections preserve important features from earlier layers. This helps maintain consistency across long information flows. Layer normalization improves convergence and prevents model collapse. These components ensure reliable performance throughout training.

3.3.4 Disaster Feature Extraction

ViT outputs high-level feature representations that summarize the characteristics of the disaster across the region. Low-backscatter regions are examined to create flood masks. Linked water spreads. Large areas of water accumulation are identified by the model. For early flood assessment, these masks are crucial.

They aid in downstream risk planning as well. The flood coverage aids

in giving priority to important areas. Maps of the likelihood of landslides are created by identifying sudden changes in the terrain and patterns of soil displacement. Landslide-prone areas can be distinguished using SAR backscatter anomalies. These discrepancies are captured by ViT on both a local and global level. Early warning systems are improved as a result. Rescue teams can concentrate on unstable slopes thanks to the forecasts.

Building collapse signatures are extracted by analyzing changes in structural radar returns. The model detects broken patterns, sudden intensity drops, and displaced structures. This helps identify regions with significant infrastructure damage. The output supports damage assessment reports. It allows efficient resource allocation.

The deformation vectors of the terrain describe the horizontal and vertical shifts detected in the landscape. These shifts are important for detecting subsidence, cracks, and ground instability. The vectors help understand deeper earth movements. They assist in post-disaster geo technical analysis. This improves hazard prediction.

Wide-area change detection compares pre- and post-disaster SAR images to identify significant variations. The model highlights regions with severe environmental transformation. Change detection supports overall disaster mapping. It helps authorities assess impact severity. This becomes a key input for emergency planning.

Output: high-level disaster embeddings with clearly detected regions, ready for downstream fusion with YOLOv8 and LangGraph agents.

3.4 Structured Disaster Representation

The ViT outputs are converted into a structured, machine-readable event format:

```
{
  "disaster_type": "Flood",
  "severity_score": 0.91,
  "affected_coordinates": [...],
  "water_depth_pattern": ...,
  "infrastructure_damage": [...],
  "confidence": 0.92
}
```

This representation serves as input to the LangGraph multi-agent system.

3.5 Human Detection via YOLOv8 on C2A Dataset

Night-time optical images from the C2A dataset [1] are processed using the YOLOv8 model to detect human presence in disaster-affected regions. Because of its strong feature extraction capabilities, the model can function in low light. It detects potential human silhouettes in noisy or partially obscured environments by examining every frame. Rescue operations are able to obtain early situational awareness as a result. In general, the detection module serves as the pipeline's initial layer of intelligence.

For each identified human instance in an image, the system generates bounding boxes and confidence scores. These confidence values prioritize high-certainty detections while assisting in the removal of false positives. People who are stranded can be precisely located on the map thanks to the bounding boxes. Compatibility with downstream reasoning agents is guaranteed by this structured output format. In the end, it aids in developing a trustworthy framework for making decisions.

The results of human detection are incorporated into LangGraph's planning agents to pinpoint critical areas that need to be addressed right away. For risk assessment, the detected coordinates are combined with environmental and hazard layers. Priority zones are dynamically updated as new frames are continuously processed by the model. This enables the system to modify rescue tactics almost instantly. The integration guarantees that real visual evidence serves as the foundation for AI-driven planning. Based on the integrated detections, rescue teams are guided to regions where stranded individuals are likely present. The system overlays detection markers on GIS maps, simplifying navigation for field forces. These visual cues help teams understand exact risk hotspots instead of relying solely on manual search efforts. Route suggestions are also aligned with detection zones for efficient task execution. This greatly increases the chances of timely rescue.

3.6 LangGraph Multi-Agent System

In order to manage reasoning, planning, routing, and communication in a coordinated pipeline, LangGraph coordinates several AI agents. Although

each agent focuses on a particular aspect, they all work well together. The system can adapt to various disaster scenarios thanks to its modular design. Additionally, it guarantees decision-making processes are transparent. The agents collaborate to create disaster response plans that are both practical and context-aware.

3.6.1 RAG (Retrieval-Augmented Generation) Agent

The RAG agent retrieves official government SOPs, infrastructure vulnerability information, casualty statistics, and historical disaster reports. In order to contextualise current observations, it also collects risk maps at the state and district levels. It creates a rich background context for the ViT and YOLOv8 outputs by combining this varied knowledge. This lowers uncertainty and enhances the quality of reasoning. Other agents can use the information that was retrieved as a knowledge pack.

To assist downstream planning agents in making well-informed decisions, the agent enhances the visual outputs with pertinent contextual information. It serves as the system's memory layer, continuously growing as new information becomes accessible. By matching recent detections with past patterns, the contextual pack facilitates pattern recognition. This increases the system's adaptability in challenging disaster situations. The recommendations' relevance and accuracy are improved by the knowledge augmentation.

3.6.2 Reasoning & Planning Agent

This agent prioritizes zones by analyzing flood severity, human detection outputs, environmental hazards, and local population density. It synthesizes

this data to derive risk levels for different regions. By performing multi-criteria reasoning, it identifies where intervention is urgently required. The agent ensures that limited rescue resources are allocated optimally. This structured prioritization significantly improves the operational efficiency of rescue missions.

Additionally, the agent determines safe routes for rescue teams to approach and estimates possible casualties. To identify safe navigation routes, it assesses surrounding hazards, water flow direction, and road conditions. Casualty estimation models are supported by historical patterns that the RAG agent retrieves. More precise planning is made possible by the integration of historical and real-time data. In the end, it lowers risks for both rescuers and victims.

The agent creates a thorough reasoning chain that explains the conclusion behind the risk score for each high-risk zone. Decision traceability and transparency are guaranteed by these reasoning chains. They make it easier for stakeholders to comprehend why some zones were given priority over others. During disaster response operations, this promotes accountability. Over time, the chains also help to improve the system.

3.6.3 Routing & Logistics Agent

Using GIS systems, mapping APIs, and terrain-aware models, the routing agent determines the quickest and safest routes. It takes into account current risks like water depth, blocked roads, and fallen buildings. If the main routes are no longer usable, alternate routes are automatically created. In an emergency, this guarantees continuous navigation assistance. As new field

data is received, the routing logic continuously adjusts its routing plans, the agent takes into account accessible road networks, boat navigation channels, and helicopter landing zones. This enables the use of multiple rescue techniques based on the severity and topography. It guarantees that the best mode of transportation is suggested in every circumstance. It improves mission flexibility by integrating air, sea, and land routes. The effectiveness of rescue operations is increased by this comprehensive routing strategy.

Structured rescue plans with estimated time of arrival (ETA), risk alerts, and suggested operational procedures are among the final products. Field teams can easily understand these plans thanks to their format. Integration with communication dashboards is made easy by its structured nature. For contingency planning, hazards and alternate routes are also included. For responders on the ground, the output becomes a crucial operational guideline.

3.6.4 Communication Agent

Structured rescue plans are transformed into human-readable formats, like reports, messages, or alerts, by the communication agent. In order to make the suggested actions easily understandable to field operators and officials, it simplifies technical information. During translation, the agent makes sure that no important information is lost. Teams can coordinate more successfully as a result. It acts as the last point of contact between human decision-makers and AI insights.

Rescue teams' dashboards and mobile applications receive alerts, updated risk maps, and GIS overlays from the agent. All team members are able to maintain situational awareness in real time thanks to these updates. When

high-risk changes take place, it guarantees that field units are immediately notified. Coordination between command centres and ground units is enhanced as a result. During emergencies, quick communication is essential to saving lives.

3.7 Rescue Team Interface

Real-time maps of the impacted areas are provided by the interface, with hazard locations, flooded areas, and detected human activity highlighted. Rescue teams can quickly grasp ground realities with the aid of this visualisation. For simpler decision-making, it provides user-friendly map overlays. As new detections come in, the interface is constantly updated. This guarantees that teams are always working with the most recent data.

Field teams can view safe route recommendations and hazard alerts directly on dashboards or mobile devices. The amount of time spent manually creating navigation routes is decreased by these insights. Deep-water pockets, blocked roads, and safe detours are highlighted on the interface. Teams can view the ETA for various routes as well. Rescue operations become much faster and safer as a result.

The system integrates YOLOv8 human detection with SAR-ViT-based flood forecasts to create a coherent operational picture. This multimodal integration allows teams to focus on areas where human presence coincides with significant flood impact. It improves the accuracy of risk assessments and mission planning. Decision-making becomes more precise and data-driven.

This ultimately increases the efficacy of rescues.

3.8 Feedback Loop

Field teams regularly update the LangGraph system with new hazards, rescued individuals, and blocked routes. These updates are automatically incorporated into the agents' reasoning pipelines. This maintains the accuracy of the system in real time. The feedback ensures that the AI will adapt to changing ground conditions. It drastically reduces recommendations that are erroneous or outdated.

New event reports are stored by the RAG agent and added to its growing knowledge base. The contextual understanding of future disasters is strengthened over time by this enriched dataset. Additionally, it enables more precise retrieval for comparable areas or kinds of disasters. This enhances the system's overall intelligence. With each mission, the memory expands dynamically.

Periodically, fresh SAR image pairs from the impacted areas are used to refine the ViT model. This enhances its capacity to forecast structural damage and flood severity in a variety of terrains. The process of fine-tuning improves generalisation for upcoming occurrences. The model's predictive performance improves as it gains knowledge from real-world situations. This guarantees that the system stays current with changing disaster conditions.

All detection and reasoning modules are made more reliable by the ongoing feedback loop. The system becomes more intelligent and context-aware with each mission. In the event of future disasters, this enables rescue teams to get more precise and timely recommendations. Long-term enhancement of the entire AI pipeline is supported by the loop. In the end, it helps make disaster response operations quicker and more efficient.

Chapter 4

Experimentation Results

The results for YOLO model on C2A dataset [1] and CNN and ViT model on SEN12-FLOOD Dataset [2] are discussed in this chapter.

4.1 Human Detection on C2A Dataset

This section presents the results of human detection using YOLOv8 on the C2A dataset [2]. The model's performance is evaluated using various metrics and visualizations.

4.1.1 Confusion Matrix

The confusion matrix shows the number of true positives, false positives, and false negatives for human detection. Fig. 4.1 shows Confusion matrix of YOLOv8 on the C2A dataset.

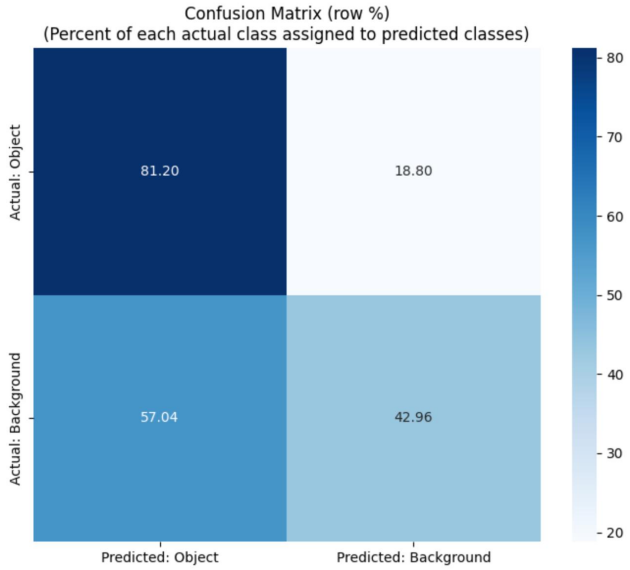

Figure 4.1: *Confusion matrix of YOLOv8 on the C2A dataset*

4.1.2 Loss Curves

The Figure 4.2, Figure 4.3, Figure 4.4 show the evolution of various graphs during training.

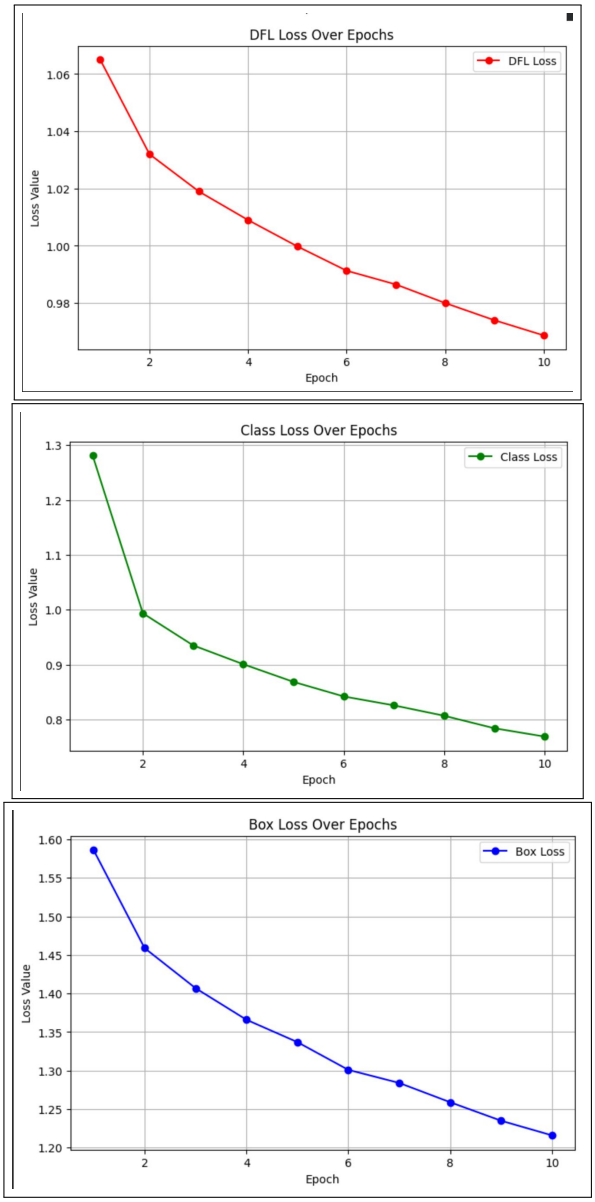

Figure 4.2: *1st row: DFL loss, 2nd row: Class loss, 3rd row: Box loss.*

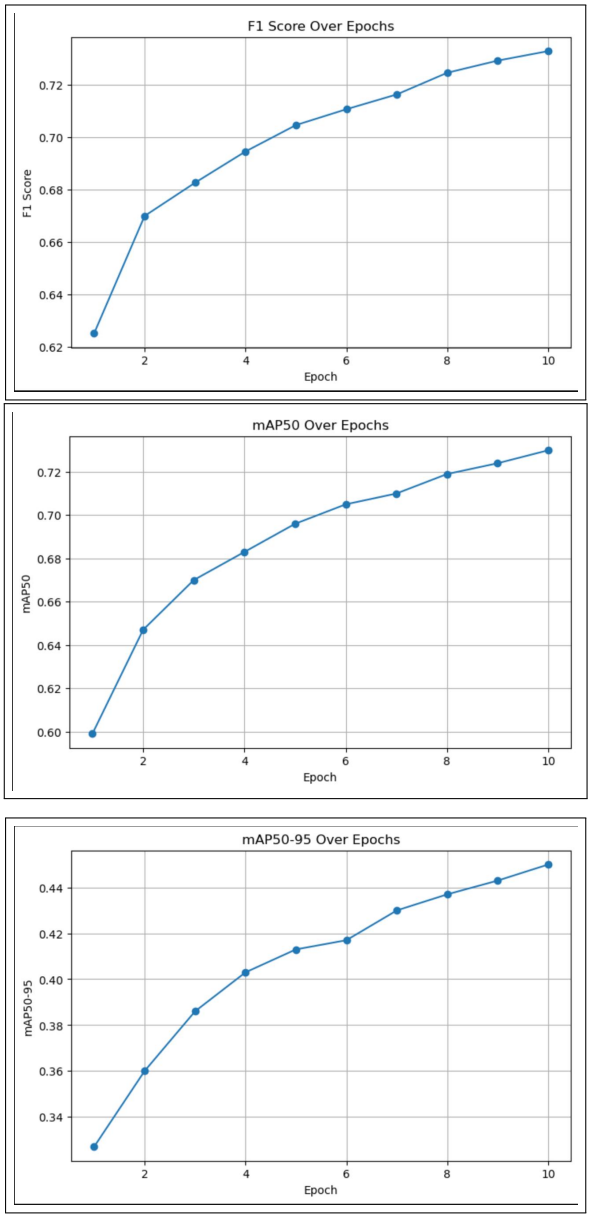

Figure 4.3: *1st row: F1-score, 2nd row: mAP@50, 3rd row: mAP@50-95.*

4.1.3 Recall and Precision

The precision-recall curves provide insight into the model's detection performance at different confidence thresholds.

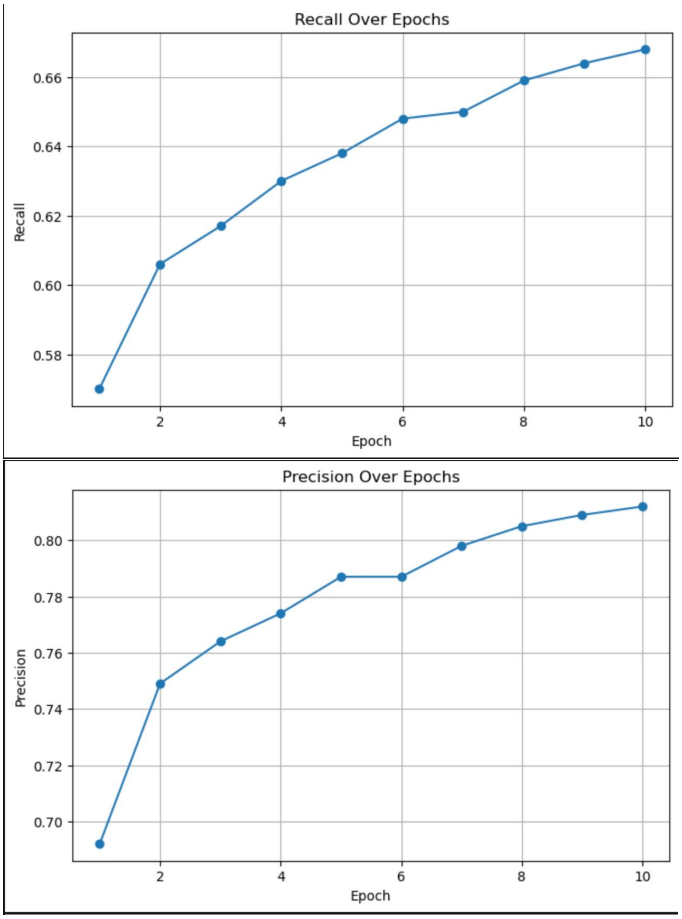

Figure 4.4: *1st row: Recall curve, 2nd row: Precision curve of YOLOv8 on C2A dataset.*

4.1.4 Summary

Overall, YOLOv8 performs effectively for human detection in night-time disaster images from the C2A dataset [1]. The detailed metrics and curves confirm that the model achieves high precision, recall, and mAP values.

4.2 Flood Classification on SEN12-FLOOD Dataset

SEN12-FLOOD Dataset has been trained on two models : CNN and ViT. Table 4.1 shows the comparative results of the two models trained on the dataset.

Table 4.1: Classification performance

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
CNN	82.0	81.5	82.0	81.7
ViT	83.8	83.9	83.8	83.8

The result for CNN-Based model on the dataset is shown in Table 4.2, Figure 4.5, Figure 4.6, Figure 4.7.

Table 4.2: CNN performance

Metric	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
CNN	82.0	81.5	82.0	81.7

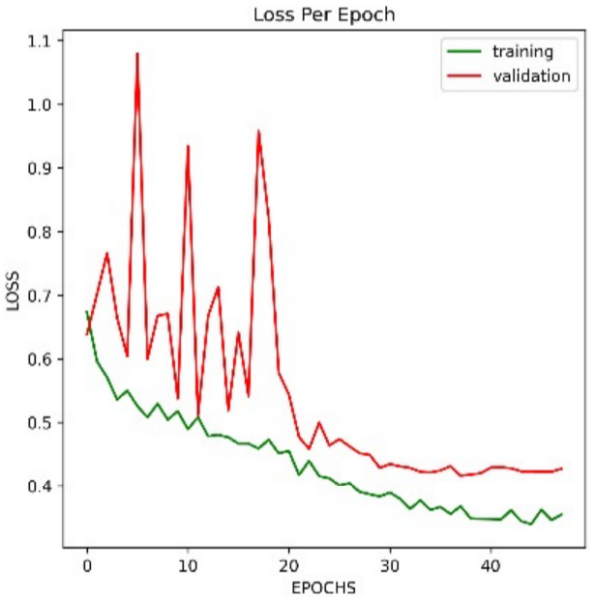

Figure 4.5: *CNN training and validation loss on the SEN12-FLOOD dataset*

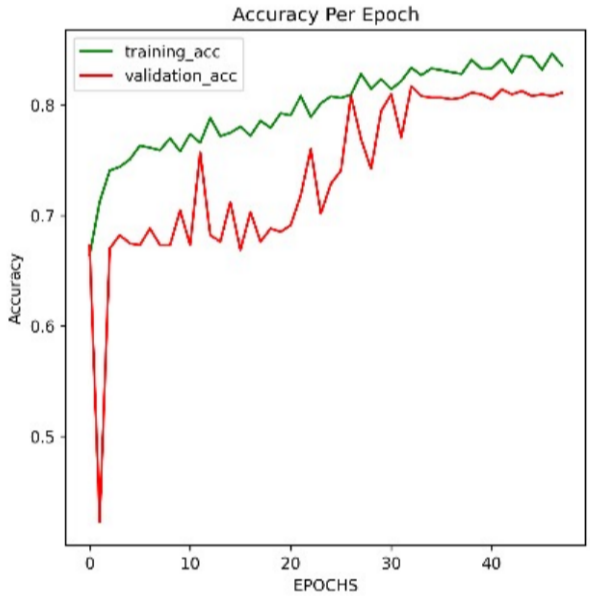

Figure 4.6: *CNN training and validation accuracy*

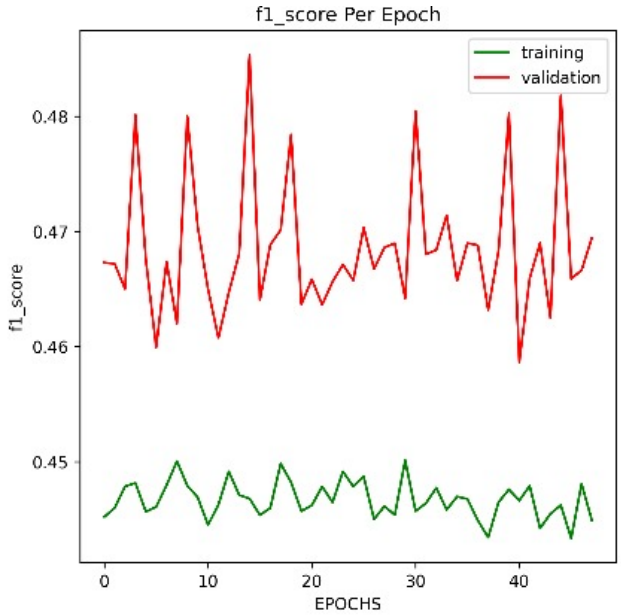

Figure 4.7: *CNN training and validation F1-score*

The result for ViT-Based model on the dataset is shown in Table 4.3,Figure 4.8,Figure 4.9

Table 4.3: ViT Classification Report

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0	0.886	0.873	0.880	339
1	0.739	0.763	0.751	160
Macro Avg	0.813	0.818	0.815	499
Weighted Avg	0.839	0.838	0.838	499
Accuracy		0.838		

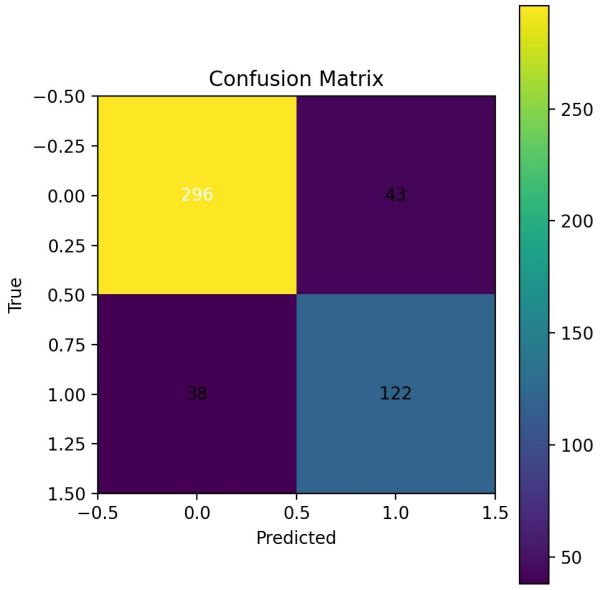

Figure 4.8: *Confusion matrix for ViT*

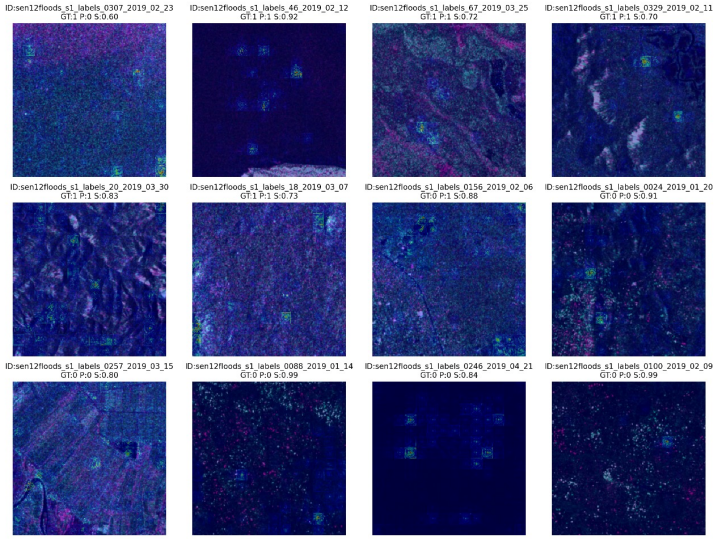

Figure 4.9: *SAR flood detection outputs*

4.3 Overall Analysis

The proposed framework demonstrates strong performance in both human detection and flood classification tasks. YOLOv8 effectively detects humans in night-time disaster images under challenging conditions, with recall and precision curves indicating balanced and stable performance. CNN achieves solid flood classification with 82% accuracy and balanced precision, recall, and F1-score, while ViT further improves performance with 83.8% accuracy and better handling of minority classes. Training curves for both CNN and ViT show smooth convergence in loss, accuracy, and F1-score, confirming stable learning. Confusion matrices and classification reports demonstrate ViT's superior generalization, and SAR flood imagery qualitatively validates model predictions with confidence scores, showing more accurate detection of flooded areas compared to CNN. Overall, the combination of YOLOv8 for human detection and CNN/ViT for flood classification provides a robust, complementary framework for disaster monitoring, integrating night-time optical imagery and SAR data to achieve reliable situational awareness.

Chapter 5

Conclusion and Future Work

A deep learning–based framework for high-resolution night-time disaster monitoring that integrates both optical and radar remote sensing data. The study focused on two key objectives: (1) detecting humans in complex night-time disaster imagery using optical data, and (2) identifying flood-affected regions from Synthetic Aperture Radar (SAR) imagery. Together, these approaches contribute to developing an automated and scalable disaster response system.

For optical imagery, the **YOLOv8** model was trained on the **C2A dataset** [1], which contains scenes of humans in various disaster contexts such as fire, flood, and building collapse. Comprehensive preprocessing steps including normalization, augmentation, and resizing were applied to extract and enhance the night-time subset of the dataset. Experimental results demonstrated that YOLOv8 effectively detects humans under low illumination, showing strong robustness to occlusions and background clutter. The model achieved a good balance between accuracy and inference speed, making it suitable for deployment in UAV- or edge-based night-time rescue systems.

For radar-based flood detection, deep learning models were trained and evaluated on the **SEN12-FLOOD dataset** [2], which provides co-registered Sentinel-1 SAR and multispectral imagery. A baseline **Convolutional Neural Network (CNN)** was implemented for binary flood classification, followed by a **Vision Transformer (ViT)** model designed to capture long-range spatial dependencies. The ViT achieved higher detection accuracy and generalization compared to the CNN, demonstrating the advantage of transformer-based architectures for radar-based disaster analysis.

Overall, the results confirm that combining high-resolution optical imagery and radar sensing data significantly enhances situational awareness in disaster environments. The proposed system successfully automated human and flood detection tasks under low-light and all-weather conditions, laying a strong foundation for real-time, AI-driven disaster monitoring and management.

While the developed framework achieved promising results, as a future work several enhancements are envisioned to further improve performance, adaptability, and real-world applicability.

- **Real-World Dataset Expansion:** Additional real and night-time UAV or satellite imagery will be collected to reduce dependency on synthetic datasets and improve model generalization to actual disaster conditions.
- **Multi-Agent Reasoning:** Implement LangGraph agents, including RAG, reasoning, planning, and communication modules, to generate actionable disaster response recommendations.

- **Real-Time Feedback:** Develop a feedback loop to incorporate field updates into SAR perception models and human detection networks, allowing adaptive learning and improved accuracy.

In summary, this work demonstrates the potential of deep learning for high-resolution, night-time disaster monitoring using both optical and radar remote sensing. By advancing toward transformer-based hybrid architectures and multi-modal data fusion, future iterations of this system aim to deliver a fully autonomous, real-time solution for efficient and safe disaster response operations.

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